



NETWORK LAYER

Module 3

Problem: An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets.

- a. Find the subnet mask.
- b. Find the number of addresses in each subnet.
- c. Find the first and the last address in the first subnet.
- d. Find the first and the last address in the last subnet (subnet 1024).

Solution: Do it in class.

Problem: An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets.

- Find the subnet mask. **Answer: 255.255.255.192/26**
- Find the number of addresses in each subnet. **Answer: 64 addresses**
- Find the first and the last address in the first subnet. **Answer: 130.56.0.0/26 – 130.56.0.63/26**
- Find the first and the last address in the last subnet (subnet 1024). **Answer: 130.56.255.192/26 – 130.56.255.255/26**

Problem: An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets.

- a. Find the subnet mask.
- b. Find the number of addresses in each subnet.
- c. Find the first and the last address in the first subnet.
- d. Find the first and the last address in the last subnet (subnet 500).

Solution: Do it in class.

Problem: An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets.

- Find the subnet mask. **Answer: 255.255.128.0/17**
- Find the number of addresses in each subnet. **Answer: 32,768 addresses**
- Find the first and the last address in the first subnet. **Answer: 16.0.0.0/17 --- 16.0.127.255/17**
- Find the first and the last address in the last subnet (subnet 500). **Answer: 16.249.128.0/17 --- 16.249.255.255/17**

Problem: An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

The first group has 64 customers; each needs approximately 256 addresses.

The second group has 128 customers; each needs approximately 128 addresses.

The third group has 128 customers; each needs approximately 64 addresses.

Find the first and last addresses of each group and also find out how many addresses are still available after these allocations.

Solution: Do it in class.

Problem: An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

The first group has 64 customers; each needs approximately 256 addresses.

The second group has 128 customers; each needs approximately 128 addresses.

The third group has 128 customers; each needs approximately 64 addresses.

Find the first and last addresses of each group and also find out how many addresses are still available after these allocations.

Answer: Group 1: 190.100.0.0/18 – 190.100.63.255/18

Group 2: 190.100.64.0/18 – 190.100.127.255/18

Group 3: 190.100.128.0/19 – 190.100.159.255/19

Number of Addresses still Available = 24576 Addresses

References:

Data Communications and Networking by Behrouz A. Forouzan.



THANK YOU